

Processing Big Data Using MongoDB-GridFS

MongoDB becomes much more efficient when combined with GridFS. The latter divides a document into parts or chunks that are then stored as separate documents. In a nutshell, GridFS is a kind of file system used to store files.



The rapidly increasing volume and variety of unstructured data generates enormous datasets which are difficult to analyse and extract further knowledge from. Issues related to the storage, processing and knowledge discovery from huge amounts of unstructured data are addressed by Big Data analytics. There are various applications for which the volume, velocity and variety of data increases very frequently, and a significant amount of research work is going on to understand how Big Data analytics could help in processing and understanding such data.

The following list gives a rough idea of why Big Data analytics has gained so much importance today:

- Since the inception of The Indian Railway Catering and Tourism Corporation or IRCTC in 2002, online ticket bookings have increased from 29 tickets to 1.3 million tickets per day.
- As per recent research at Harvard, the information stored in 1 gram of DNA is equivalent to 700 terabytes of data in digital format.
- According to *InternetLiveStats.com*, around 10,000 tweets are processed per second on Twitter; the Internet traffic per second totals more than 42TB; and on YouTube, more than 68,000 videos are viewed every second.
- As per a report on popular statistics portal *Statista.com*, there were more than 1000 million users per day on Facebook, as of December 2016.

MongoDB: A prominent NoSQL database for Big Data processing

MongoDB is one of the widely used NoSQL databases under free and open source distribution. It is a document-oriented database written in C++, and a leader in the database segment for large scale as well as performance-aware Big Data based applications. The implementation and storage of Aadhaar cards in India is being effectively done with the integration of MongoDB.

MongoDB provides a number of modules and specifications to support scalable and high performance Big Data based applications. These features and modules include GridFS, Sharding, Capped Collections, Map Reduce and many others. MongoDB can insert more than 200 million rows of data in a few seconds.

GridFS—for the storage and retrieval of large objects

GridFS is one of the powerful specifications of MongoDB that